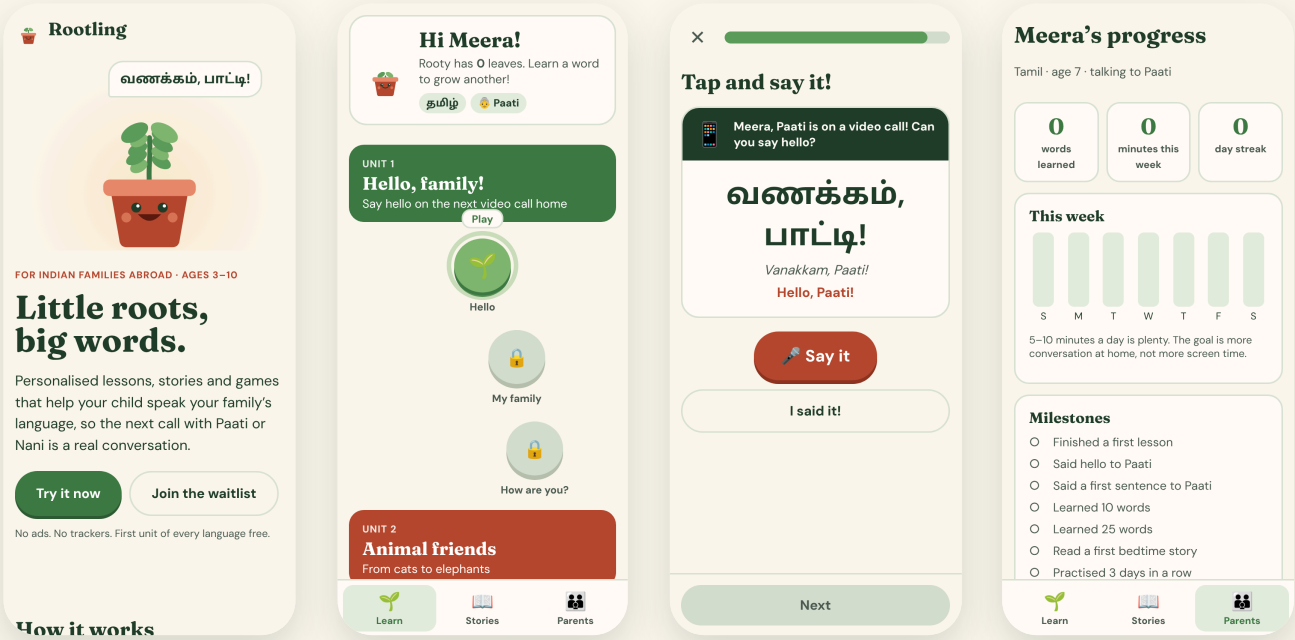
PROJECT SUBMISSION · NOVEMBER 2026

Rootling

Little roots, big words.

A web app that helps children aged 3–10 in Indian families abroad learn to speak their family's language (Tamil, Telugu, Punjabi, Gujarati, Malayalam) through personalised lessons, stories and "say it to family" moments.



Live app	**https://www.rootling.space**
Source code	**github.com/aakrishtkalra1999-ux/skills-installation/tree/main/rootling-app**
Submitted by	_____
Course / assignment	_____

All links

WHAT	LINK
Live app	**https://www.rootling.space** Works on any phone or laptop browser. Best on a phone.
Source code	**https://github.com/aakrishtkalra1999-ux/skills-installation/tree/main/rootling-app**
Product spec (PRD)	**https://github.com/aakrishtkalra1999-ux/skills-installation/blob/main/rootling-app/PRD.md**
Build & deploy guide	**https://github.com/aakrishtkalra1999-ux/skills-installation/blob/main/rootling-app/SETUP_AND_DEPLOY_GUIDE.md**
Database schema & security rules	**https://github.com/aakrishtkalra1999-ux/skills-installation/blob/main/rootling-app/supabase/schema.sql**
Native-speaker review sheet	**https://github.com/aakrishtkalra1999-ux/skills-installation/blob/main/rootling-app/content/review.csv**
Privacy policy (draft)	**https://www.rootling.space/privacy**
Terms of use (draft)	**https://www.rootling.space/terms**
Main build (pull request)	**https://github.com/aakrishtkalra1999-ux/skills-installation/pull/1**
Bug fix after launch	**https://github.com/aakrishtkalra1999-ux/skills-installation/pull/2**

How to try it (5 minutes)

Demo parent login: email _____ password _____

Or create your own parent account with any email you can open (a confirmation link is emailed).

1. Open **www.rootling.space**. The landing page explains the product and has a **waitlist** form.
2. Press **Start free** → create a parent account, or **Log in** with the demo login above.
3. Tick the **parent consent** box → choose a language → enter a child's first name and age → pick favourite topics and the family member they talk to (e.g. Paati, Ammamma).
4. Tap the green **Play** lesson. Try an age under 6 (pictures and sounds only) and over 6 (reading and sentence building).
5. Open **Stories** for a bedtime story that stars the child and their grandparent.
6. Open **Parents** and answer the grown-ups sum: progress, milestones, words to practise, settings, family, plan and account.
7. Under **Your plan** press **Start your free month**. Payments are in **Stripe test mode**: use card `4242 4242 4242 4242`, any future date, any CVC. No real money is charged. All lessons then unlock.

Good to know. Word audio uses the device's built-in voice until native-speaker recordings are added, so some phones (often iPhones) have no Tamil or Telugu voice; lessons still work with pictures and text. The "Say it" microphone needs Chrome; elsewhere children tap "I said it!".

1 The problem

Millions of Indian families live in the US, UK, Canada, Australia, Singapore and the Gulf. Their children often grow up understanding a little of the family language but replying in English. Calls with grandparents become one-word answers, and parents who are busy (or not fluent themselves) struggle to teach it.

Existing language apps target adults, focus on Hindi or major world languages, assume the learner can read, and use streaks and penalties designed for grown-ups. Little exists for a 4-year-old learning *Tamil to talk to Paati*.

Who it is for

USER	NEED	HOW ROOTLING SERVES THEM
Parents (customer, account owner)	A safe, ad-free way to pass on the language, and to see progress	Parent-owned account, consent first, a parents area with progress and words to practise at home
Children 3–10 (learner)	Short, playful lessons that work before they can read	Picture-and-sound exercises, no lives or penalties, a mascot that grows with them
Grandparents (audience)	Real conversation with grandchildren	Lessons practise the exact words for the next video call, using the family's own word for grandma or grandpa

2 The solution

Rootling is an installable web app (it works in any browser and can be added to a phone's home screen). A parent sets it up, and the child plays 5–10 minute lessons that end in a "Tap and say it" moment: a video-call scene where the child says a real line to their grandparent.

5	28	107	5
languages	lessons	words	bedtime stories

LANGUAGE	STATUS	UNITS	LESSONS	WORDS	SAY-IT LINES	STORIES
Tamil (தமிழ்)	Launch	5	8	27	4	1
Telugu (తెలుగు)	Preview	5	8	28	4	1
Punjabi	Preview	3	4	18	2	1
Gujarati	Preview	3	4	17	2	1
Malayalam	Preview	3	4	17	2	1

"Preview" languages are complete in the app but wait for a native speaker to sign off every line (see the review sheet link).

3 What the app does

Landing page and waitlist

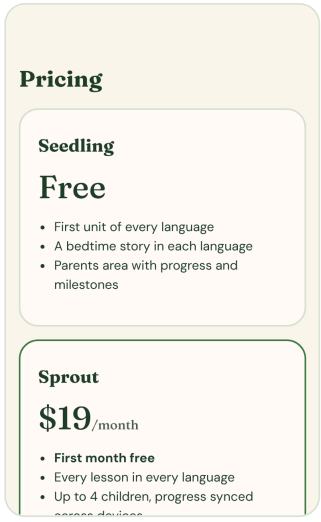
Explains the product to parents (how it works, Rooty, languages, pricing, privacy promise, FAQ). The waitlist records email, language wanted and country, so the most-requested language is built next.



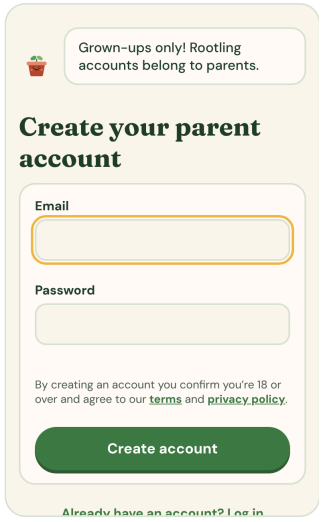
Landing page hero



How it works



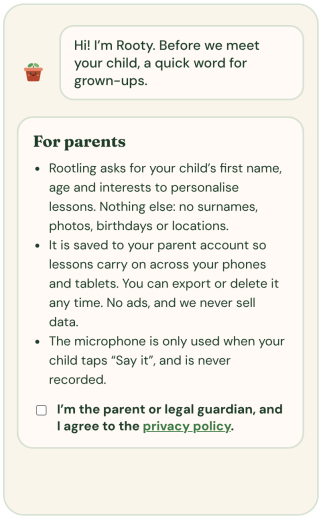
Free Seedling plan and $19 Sprout plan



Parent sign-up (children never log in)

Parent-led setup

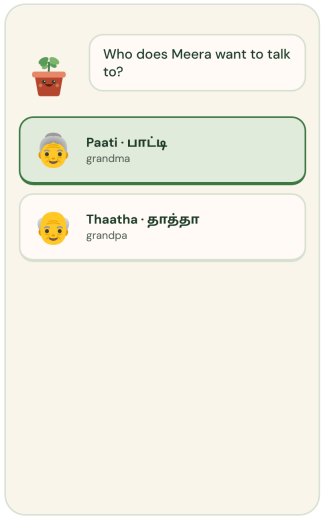
The parent agrees first, then picks the language, the child's first name and age, favourite topics (which move earlier in the path) and the family member the child will talk to. That word (Paati, Thaatha, Ammamma…) appears throughout lessons and stories.



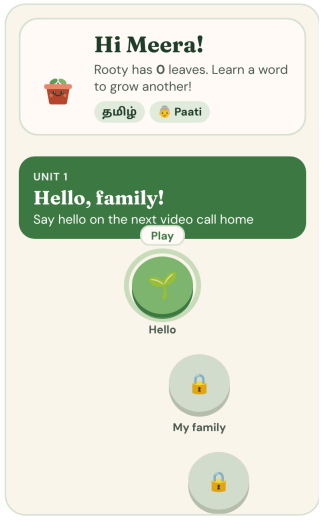
Consent before any child details



Choose the family language



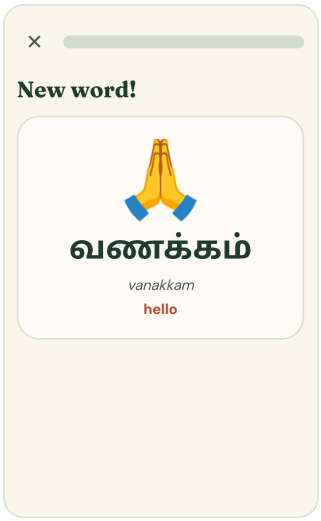
Who the child talks to



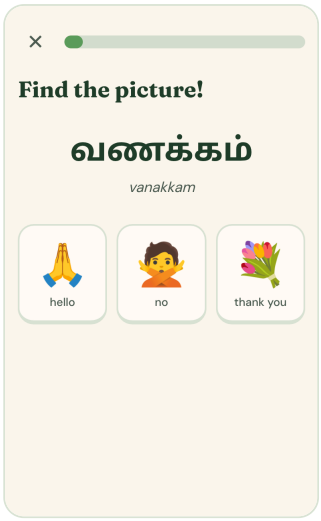
The lesson path; Rooty grows a leaf per word

Lessons built for small children

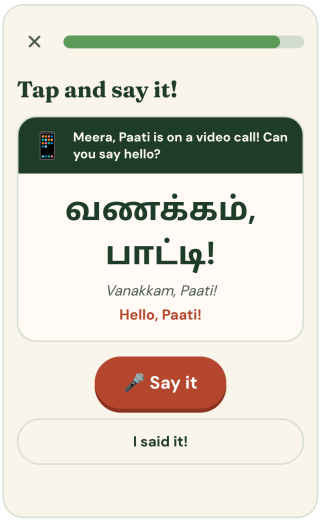
- **Every lesson ends in success.** No hearts or lives: a wrong picture fades away and the child tries again.
- **Age-aware.** Under-6s get picture and sound exercises only; older children also read words and build sentences.
- **Tap and say it.** A video-call scene (“Meera, Paati is on a video call! Can you say hello?”). The browser’s speech recognition accepts any clear try; nothing is recorded. An “I said it!” button works everywhere.



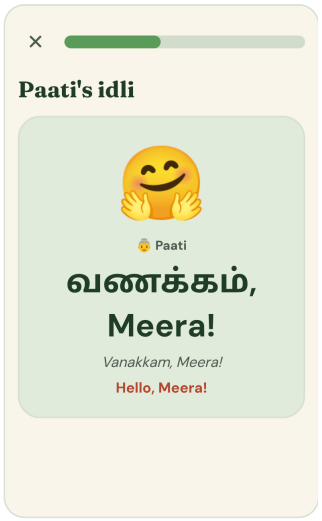
Meet a new word



Find the picture



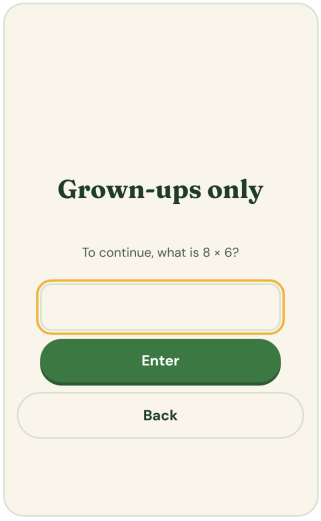
Tap and say it to Paati



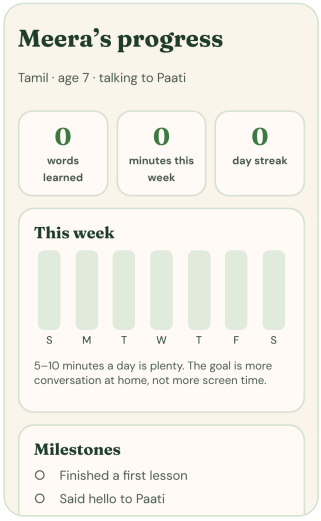
Bedtime story starring the child

Parents area, families and the Sprout plan

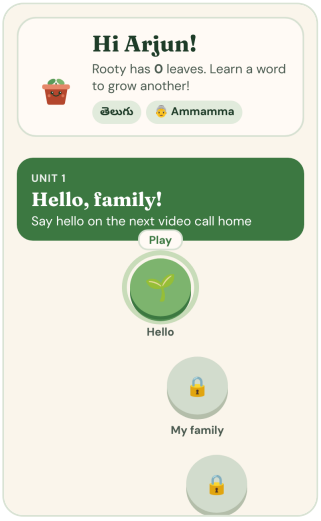
Behind a grown-ups check (a times-table sum): words learned, minutes this week, streak, milestones such as “Said hello to Paati”, words to practise at home, settings, up to 4 children per family, the subscription, data download and account deletion. The free plan opens the first unit of every language; the Sprout plan ($19/month, first month free) opens everything.



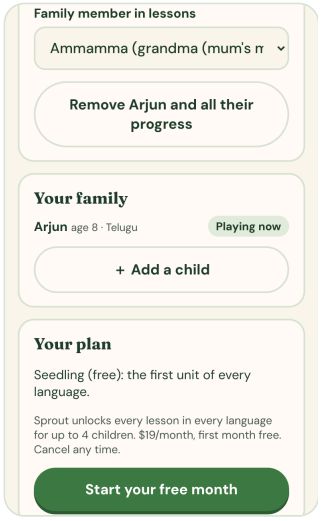
Grown-ups-only check



Progress and milestones

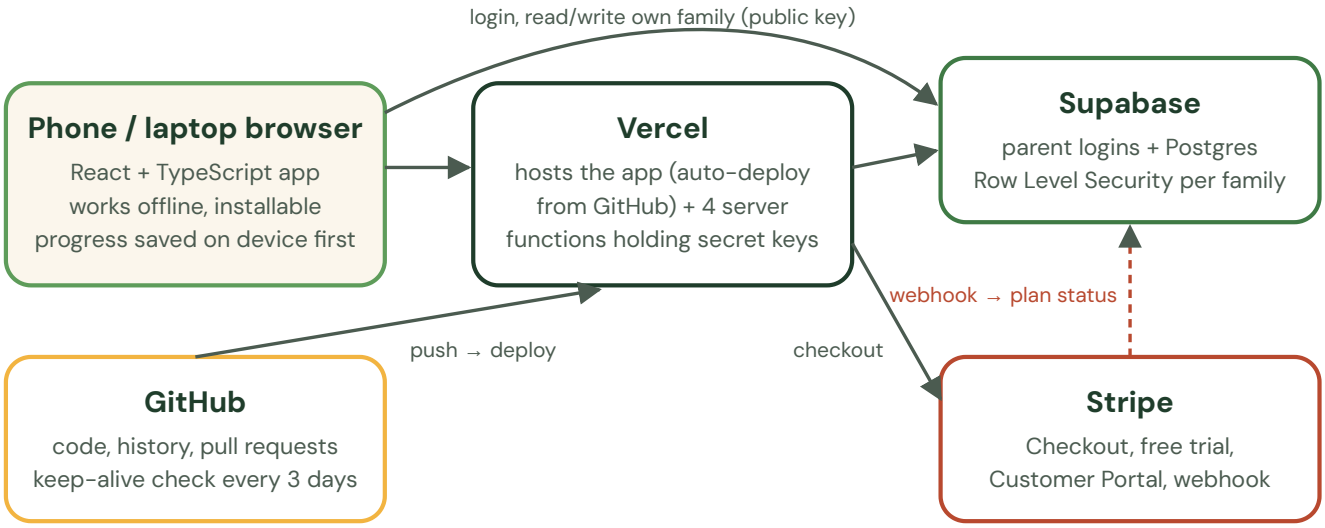


Free plan: later units wait for a grown-up



Family, plan and account

4 How it is built



LAYER	CHOICE	WHY
App	React 19, TypeScript, Vite; installable web app with a service worker	One codebase for phones, tablets and laptops; no app-store approval needed to launch
Accounts & data	Supabase Auth (email confirmation) and Postgres with Row Level Security	Each parent can only ever read or write their own family, enforced inside the database
Sync	Offline-first: saved on the device instantly, uploaded in the background, merged across devices	Children play on patchy Wi-Fi; nothing learned on a tablet is lost when they switch to a phone
Payments	Stripe Checkout + Customer Portal + webhook, in Vercel server functions	Card details never touch Rootling; secret keys never reach the browser
Hosting	Vercel, auto-deploying every push to `main`	Free to start, HTTPS, preview links for every change
Audio	Native-speaker recordings when present, device text-to-speech otherwise	Many phones lack Indian-language voices; recordings fix that without code changes

About 6,400 lines of code across the app, 4 server functions, the database schema and build scripts.

5 Child safety and privacy by design

Because the users are children, privacy rules were set before any code (US COPPA and the UK Children's Code inform them):

- **Accounts belong to parents.** Consent is recorded before any child details are entered. Children never type an email or password.
- **Data minimisation.** Per child only a first name, age, language, chosen family member, favourite topics and learning progress. No surnames, photos, birthdays, locations or voice recordings.
- **No ads, analytics or third-party trackers.** Fonts are bundled with the app, and a strict Content-Security-Policy blocks outside scripts. A browser test confirmed **0 requests to third parties** while a child plays.
- **Parents stay in control.** Download all family data as a file, remove a child, or delete the whole account (which also cancels the subscription immediately).
- **Payments only behind the grown-ups check**, never on child screens.

6 Quality and testing

CHECK	RESULT
Automated tests (content, lesson builder, personalisation, family sync, review sheet, Stripe mapping, no external URLs)	40 / 40 pass
Database security rules, run against Postgres: one parent cannot see or change another family; max 4 children; waitlist write-only; only the server writes plans; deleting an account removes everything	All pass
Supabase security advisor on the live project	0 findings
Headless-browser walkthroughs at phone size, with and without accounts	Pass
Live test on the deployed site from a real laptop: sign-up, email confirmation, consent, child, progress saved to the cloud	Pass

A real bug found and fixed after launch. On the first live test from an iPhone, the app froze after login. The database logs showed the phone re-loading the same data dozens of times a second. The cause was a React function being re-created on every screen update, which re-triggered the data load each time. It was reproduced in an automated browser (2,988 loads in 3 seconds), fixed, and verified (4 loads; consent, child and progress all saved), then shipped the same day (**pull request #2**).

7 Running it

- **Deploys:** every merge to `main` on GitHub is built and published by Vercel within about a minute.
- **Keep-alive:** Supabase pauses free projects after a week without use, so a GitHub Action reads the database every 3 days, and a scheduled check every 5 days restores the project if it has paused.
- **Costs today:** $0 (free tiers of Vercel, Supabase and Stripe test mode).

8 Business model

PLAN	PRICE	INCLUDES
Seedling	Free	First unit of every language, a bedtime story per language, the parents area
Sprout	$19 / month, first month free (once per family)	Every lesson in every language, up to 4 children, progress synced across devices, cancel any time

The waitlist records which language each family wants, so new languages are built in order of real demand: Tamil and Telugu first, then Gujarati and Punjabi, then Malayalam or Bengali.

9 Limitations and next steps

TODAY	NEXT STEP
Content written for the prototype; Telugu, Punjabi, Gujarati and Malayalam are in preview	Native-speaker review of all 179 lines in the review sheet, then mark each language “launch”
Audio uses the device voice, missing for some languages on some phones	Record native-speaker audio (a recording guide and file list are ready); Tamil first
Privacy policy and terms are drafts	Legal review for COPPA (US) and the Children’s Code (UK) before real families sign up
Payments in Stripe test mode; free hosting tiers	Switch Stripe to live mode; move to Vercel Pro and a paid Supabase plan before charging
Web app only	Wrap for the App Store and Google Play with Capacitor
No video yet	Personalised short videos with the child’s name and recorded native audio

Summary. Rootling is a working, deployed product: a landing page with a waitlist, parent accounts with consent, personalised lessons and stories in five Indian languages, cloud sync across devices, a paid plan with a free trial, and privacy protections designed for children. It is tested, live at **www.rootling.space**, and documented so it can be run and extended by someone new.